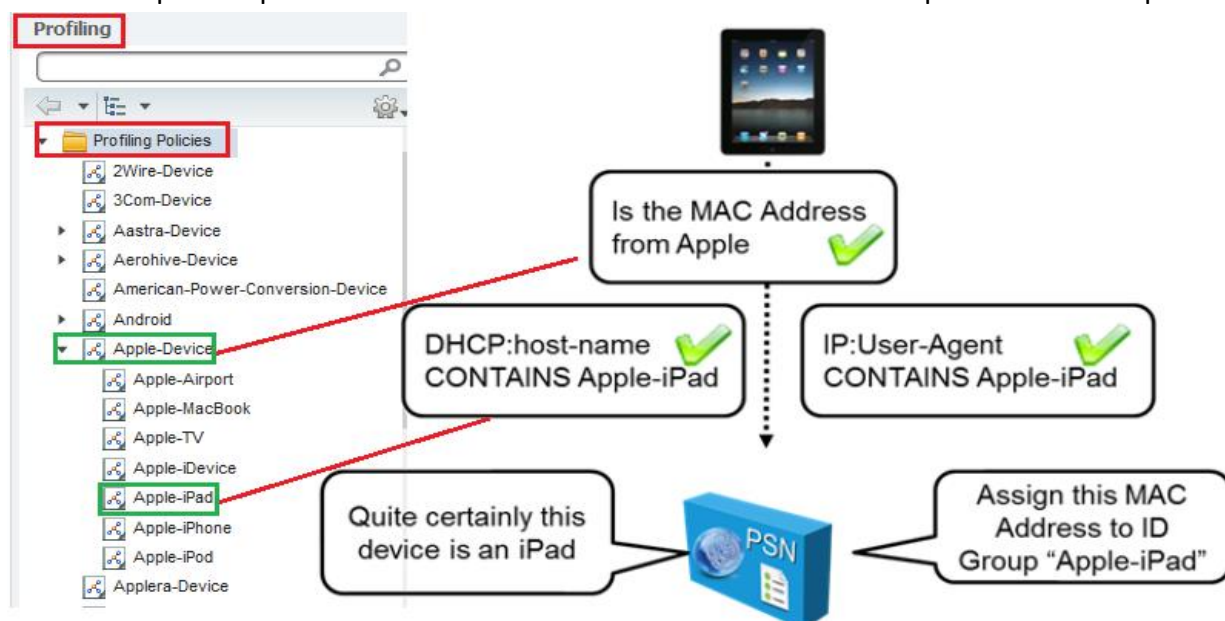


## Cisco ISE Profiler:

- o Profiling Services provides dynamic detection & classification of endpoints.
- o Profiler is responsible for endpoint detection & classification connected to network.
- o Profiling service identifies devices that connect to network and their location.
- o Profiler Identifies locates & determines capabilities of attached network endpoints.
- o Profiler using a probe or probes that collect attributes about an endpoint.
- o Profiler compares collected attributes to predefined device signatures to locate a match.
- o These predefined devices signatures are commonly, referred to as profiles.
- o Profiler facilitates effective deployment & ongoing management of authentication.
- o Endpoints are profiled, based on the endpoint profiling policies configured in Cisco ISE.
- o Profiling feature is very important in authentication & authorization of network access.
- o Profiling collects data of Endpoint devices accessing network & identifies its location.
- o Depend upon Endpoint profile; authentication server permits access of the resources.
- o Profiling service can identify, locate, and ensure the access of all endpoints connected.
- o Profiling collects attribute & classifies these endpoints into groups & stores in database.
- o Two main functions of profiler service is create and maintain learned inventory.
- o Profiler service assign endpoint identity group used in authorization policy.
- o Profiler service classify endpoints using variable sets of attributes like MAC address.
- o Profiler service using many other attributes to identify and classification endpoints.
- o Profiler service discovers & determines capabilities of endpoints attached to network.
- o Profiling requires Change of Authority to change the authorization state of an endpoint
- o Profiling policies allow categorizing discovered endpoints on network assign to groups.
- o Cisco Identify Service Engine comes with numerous built-in profiler conditions and policies.
- o Built-in profiler policies defined in hierarchical fashion & consist of parent and child policies.



## Implement Profiler Services:

Enable the profiler service on a node. Navigate to **Administration > System > Deployment > General Settings** and check **Enable Profiling Services**

This service is enabled by default and is not configurable when in standalone mode.

The screenshot displays the Cisco ISE Administration web interface. The top navigation bar includes 'Identity Services Engine', 'Home', 'Operations', 'Policy', 'Guest Access', and 'Administration' (highlighted with a red box and labeled 1). Below this, the 'System' menu is expanded, showing 'Deployment' (highlighted with a red box and labeled 3), 'Licensing', 'Certificates', 'Logging', 'Maintenance', 'Upgrade', 'Backup & Restore', and 'Admin Access'. The 'Deployment' section is further expanded, showing 'Deployment' (highlighted with a red box and labeled 4) and 'PAN Failover'. The 'Edit Node' page for 'ISE1' is shown, with the 'General Settings' tab selected (highlighted with a red box and labeled 6). The 'Personas' section lists 'Administration' (Role: STANDALONE), 'Monitoring' (Role: PRIMARY), and 'Policy Service'. The 'Enable Session Services' checkbox is checked. The 'Include Node in Node Group' dropdown is set to 'None'. The 'Enable Profiling Service' checkbox is checked (highlighted with a green box and labeled 7). A red arrow points from the text 'Enable by default in standalone' to the 'Enable Profiling Service' checkbox.

Deployment Nodes List > ISE1 5

Edit Node

General Settings 6 Profiling Configuration

Hostname ISE1

FQDN ISE1.google.com

IP Address 192.168.80.250

Node Type Identity Services Engine (ISE)

Personas

☒ Administration Role STANDALONE

☒ Monitoring Role PRIMARY

☒ Policy Service

☒ Enable Session Services ⓘ

Include Node in Node Group None

☒ Enable Profiling Service 7

Enable by default in standalone

## Implement Probes:

- o Probe is a method used to collect attribute or a set of attributes from endpoint on network.
- o By the help of Probe, Profile service collects an attribute or attributes of any endpoint.
- o Probe is software designed to collect data to be used in a profiling decision.
- o By the help of Probe, Profile service create update or modify profile in database.
- o Different Probes are responsible for collection of different type of Endpoint attributes.
- o There are nine probes on each Policy Service Node NETFLOW, DHCP, DHCPSPAN, HTTP, RADIUS, NETWORK SCAN (NMAP), DNS, SNMPQUERY and SNMPTRAP.

Navigate to **Administration > System > Deployment** > Select the **Profiling Configuration** tab  
Click the check box next to the probes want to enable.

**Deployment Nodes List > ISE1**

**Edit Node**  
General Settings | **Profiling Configuration**

| Probe                 | Enabled                             |
|-----------------------|-------------------------------------|
| 1 NETFLOW             | <input checked="" type="checkbox"/> |
| 2 DHCP                | <input checked="" type="checkbox"/> |
| 3 DHCPSPAN            | <input type="checkbox"/>            |
| 4 HTTP                | <input type="checkbox"/>            |
| 5 RADIUS              | <input checked="" type="checkbox"/> |
| 6 Network Scan (NMAP) | <input checked="" type="checkbox"/> |
| 7 DNS                 | <input type="checkbox"/>            |
| 8 SNMPQUERY           | <input checked="" type="checkbox"/> |
| 9 SNMPTRAP            | <input type="checkbox"/>            |

## NetFlow Probe:

- o Just enabling NetFlow in infrastructure and forwarding it all to the ISE.
- o It is recommended to perform extensive planning prior to use NetFlow probe.
- o Enabling check box next to the NetFlow probe & selecting Gigabit 0 interface.

**Edit Node**  
General Settings | **Profiling Configuration**

☒ **NETFLOW**

Interface: GigabitEthernet 0  
Port: 9996  
Description: The Netflow probe collects Netflow packets sent to it from Routers.

### DHCP Probe:

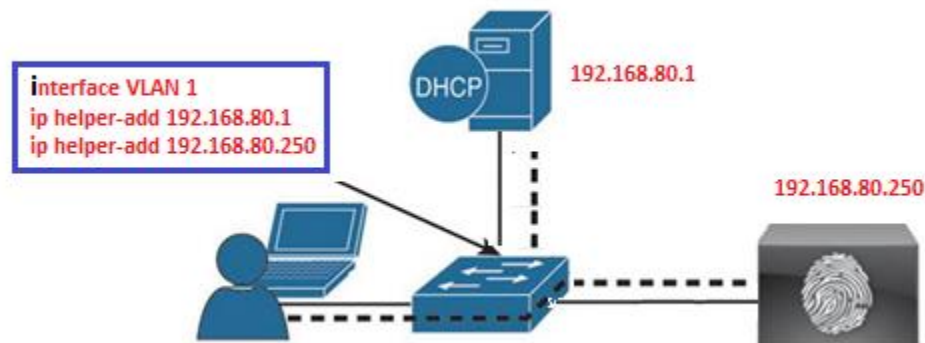
- o Configuring DHCP Packets forwarding directly to the Cisco ISE.
- o Primary use of DHCP in profiling is to capture the device MAC address.
- o DHCP requests also carry User-Agent field that helps to identify OS of device.
- o The DHCP probe requires the DHCP requests to be sent directly to the ISE.
- o Using ip helper-address interface configuration command to send request to ISE.

☒ **DHCP**

InterfaceGigabitEthernet 0

Port67

DescriptionThe DHCP probe listens for DHCP packets from IP helper.



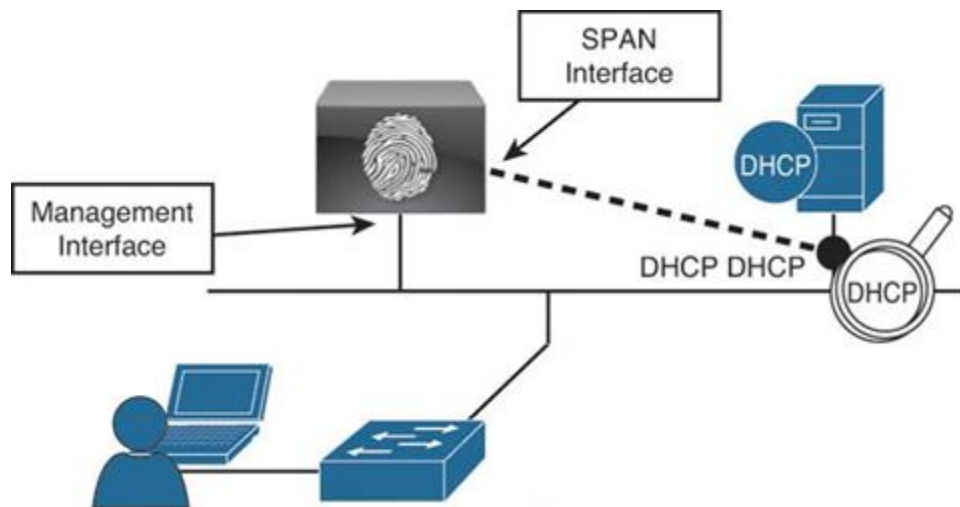
### DHCP SPAN Probe:

- o DHCP SPAN Probe is not possible or required DHCP Relay agent to configure.
- o SPAN session copies all traffic to/from source interface on a switch to a destination.
- o DHCP Helper option is more preferred than SPAN because it has less traffic overhead.

☒ **DHCPSPAN**

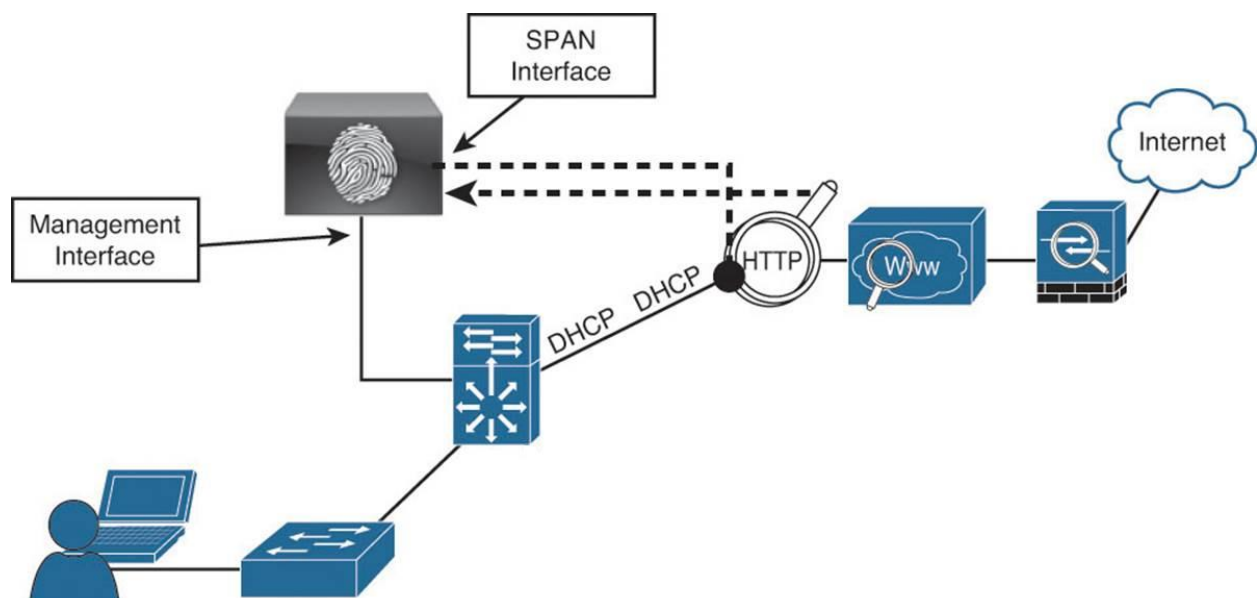
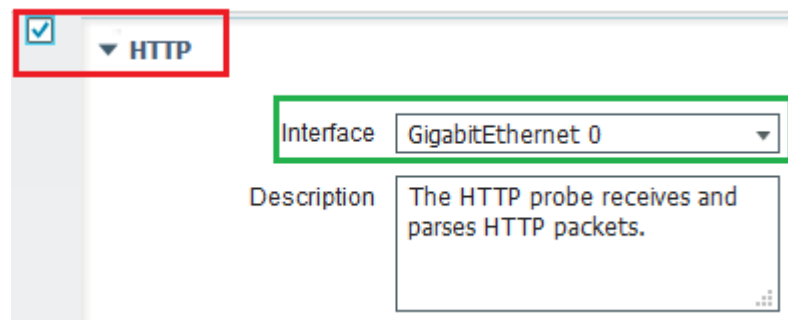
InterfaceGigabitEthernet 0

DescriptionThe DHCP span probe collects DHCP packets.



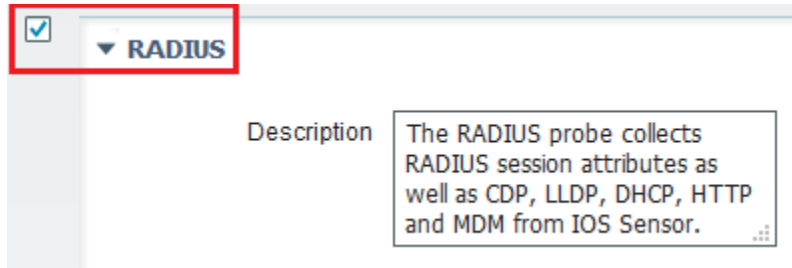
### HTTP Probe:

- o Information is transmitted in HTTP request-header field called User-Agent field.
- o The Cisco Identity Services Engine uses the information in HTTP packets.
- o User-Agent field, to help match signatures of what profile a device belongs in.
- o Use a Switched Port Analyzer (SPAN) session in true promiscuous mode.



### RADIUS Probe:

- o Calling-Station-ID field in RADIUS packet provides the endpoint's MAC address.
- o Framed-IP-Address field provides its IP address in the RADIUS accounting packet.
- o RADIUS probe in Profiling is use to Collect attributes from the RADIUS Attributes.
- o RADIUS Probe also collects other information from accounting packet of RADIUS.
- o RADIUS Probe also collects other information like CDP, LLDP and DHCP attributes.

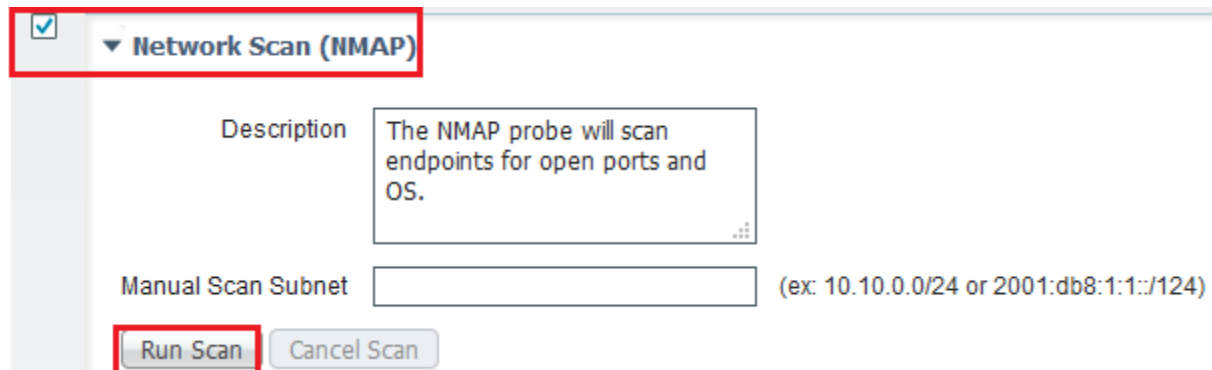


☒ ▼ RADIUS

Description The RADIUS probe collects RADIUS session attributes as well as CDP, LLDP, DHCP, HTTP and MDM from IOS Sensor.

### Network Scan (NMAP) Probe:

- o Endpoint Scanning (NMAP) probe is executed against an IP Address or subnet.
- o NMAP is a tool uses port scans, to identity a device's OS or other attributes of device.
- o Also, can run manual scan against a single node, or an entire network.
- o NMAP Probe scan endpoints for open ports and Operating System to get information.



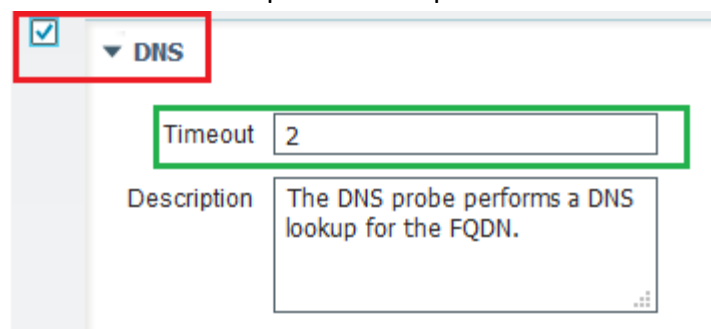
☒ ▼ Network Scan (NMAP)

Description The NMAP probe will scan endpoints for open ports and OS.

Manual Scan Subnet  (ex: 10.10.0.0/24 or 2001:db8:1:1::/124)

### DNS Probe:

- o DNS probe is used to collect the fully qualified domain name (FQDN) of endpoint.
- o Useful looking for a specific DNS name format of assets Active Directory members.
- o DNS probe require anyone from these probe DHCP, DHCP SPAN, HTTP, RADIUS, or SNMP.
- o This allows DNS probe in the profiler to do a reverse DNS lookup against specified name.



☒ ▼ DNS

Timeout

Description The DNS probe performs a DNS lookup for the FQDN.

### SNMPQUERY Probe:

- o SNMP Query Probe send query packets known as SNMP GET Requests.
- o System Queries are periodic depending upon polling interval.
- o System Queries collect Bridge, IP CDP Cache Entry LLDP Local System Data.
- o Interface Queries are generated when RADIUS Accounting Starts.
- o Interface Queries are generated when SNMP detect linkup.

☒ **SNMPQUERY**

|              |                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------|
| Retries      | <input type="text" value="2"/>                                                                    |
| Timeout      | <input type="text" value="1000"/>                                                                 |
| EventTimeout | <input type="text" value="30"/>                                                                   |
| Description  | <div>This probe collects details from network devices such as Interface, CDP, LLDP and ARP.</div> |

### SNMP Trap Probe:

- o SNMP Trap receives information from configured NAD support MAC notification.
- o SNMP Trap receives info form configured NAD support linkup, linkdown & informs.
- o For SNMPTRAP to be functional, you must also enable the SNMPQUERY probe.
- o To make this feature functional, configure the NAD to send SNMP traps or informs.
- o SNMP Trap probe receives information from the specific network access devices.
- o When ports up or go down & endpoints disconnect from or connect to network.

☒ **SNMPTRAP**

|                 |                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------|
| Link Trap Query | <input checked="" type="checkbox"/>                                                              |
| MAC Trap Query  | <input checked="" type="checkbox"/>                                                              |
| Interface       | <input type="text" value="GigabitEthernet 0"/>                                                   |
| Port            | <input type="text" value="162"/>                                                                 |
| Description     | <div>This probe receives Linkup, Linkdown and MAC notification traps from network devices.</div> |

## Profiler Feed Service:

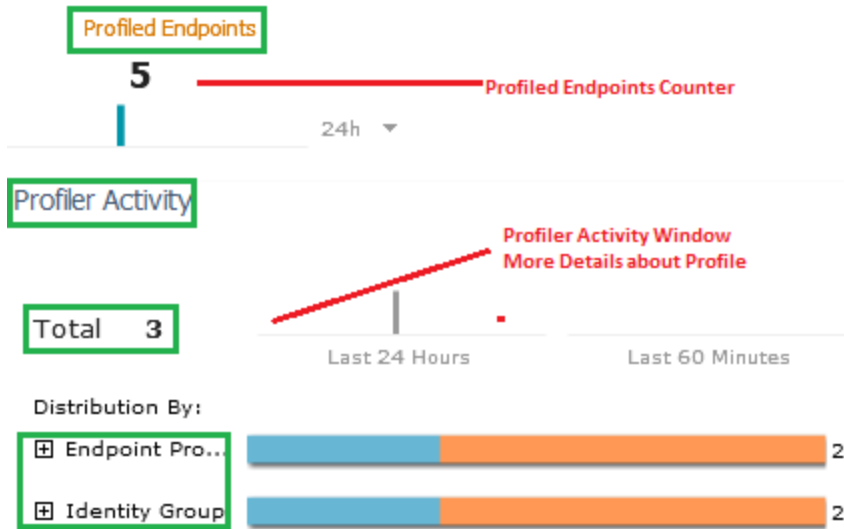
- o ISE receive updates for new, updated endpoint profiling policies and MAC OUI.
- o Cisco ISE use a feed from a designated Cisco feed server to pull profiling updates.
- o Enabling the profiler feed service allows Cisco ISE to check for new & updated profiles.
- o The profiler feed service, that you can enable will download new profiles from Cisco.
- o By default, profiler feed service is disabled, it requires Advance license to enable it.
- o In order to get updated profiles, enable the Profiler Feed Service so get new profiles.

To enable this service, navigate to **Administration>Feed Service>Profiler** and check the box next to **Enable Profiler Feed Service**. Click Save when done:

The screenshot shows the Cisco Identity Services Engine (ISE) Administration console. The top navigation bar includes 'Administration', 'Feed Service', and 'Profiler'. The 'Profiler' link is highlighted with a red box and labeled '3'. The 'Feed Service' link is also highlighted with a red box and labeled '2'. Below the navigation bar, the 'Test Profiler Feed Service Connection' section is visible. The 'Profiler Feed Service Configuration' section shows a checkbox labeled 'Enable Profiler Feed Service' which is unchecked, highlighted with a red box and labeled '4'. A red note next to it says 'Check the box to enable'.

## Verify Profiling:

- o There are many places to verify profiling operation one of them is dashboard.
- o Dashboard are numerous places to look to see whether profiling is working.
- o Counter shows number of endpoints that have been profiled in the last 24 hours.
- o Profiler Activity Window shows more detail than the Profiled Endpoints Counter.
- o Profiler Activity Window shows two different counters last 60 minutes & 24 hours.





## Implement CoA:

- o Profiling requires CoA to change the authorization state of an endpoint.
- o Cisco ISE allows a global configuration to issue a Change of Authorization (CoA).
- o Enables profiling service with more control over endpoints already authenticated.
- o After profiler CoA is enabled globally, CoA is automatically sent for any endpoint.
- o CoA sent for any endpoint that transitions from unknown to any known profile.
- o **No CoA (default)**-Use this option to disable the global configuration of CoA.
- o **Port Bounce**-Use this option, if the switch port exists with only one session.
- o Port Bounce CoA shuts down switch port & performs no shutdown to reenale it.
- o **Reauth**-Use this option to enforce reauthentication of already authenticated endpoint.
- o If the port exists with multiple sessions, then use the Reauth CoA option.
- o Change of Authorization (CoA) can be enable Per-profile as well.

To modify or change this service, navigate to **Administration>Setting>Profiling**

The screenshot shows the Cisco Identity Services Engine (ISE) Administration console. The navigation pane on the left has 'Profiling' highlighted. The main content area is titled 'Profiler Configuration'. It contains several fields: '\* CoA Type:' with a dropdown menu showing 'No CoA', 'No CoA', 'Port Bounce', and 'Reauth'; 'Current custom SNMP community strings:'; 'Change custom SNMP community strings:'; and 'Confirm changed custom SNMP community strings:'. A 'Show' button is next to the community strings fields. The 'No CoA' option is selected in the dropdown.

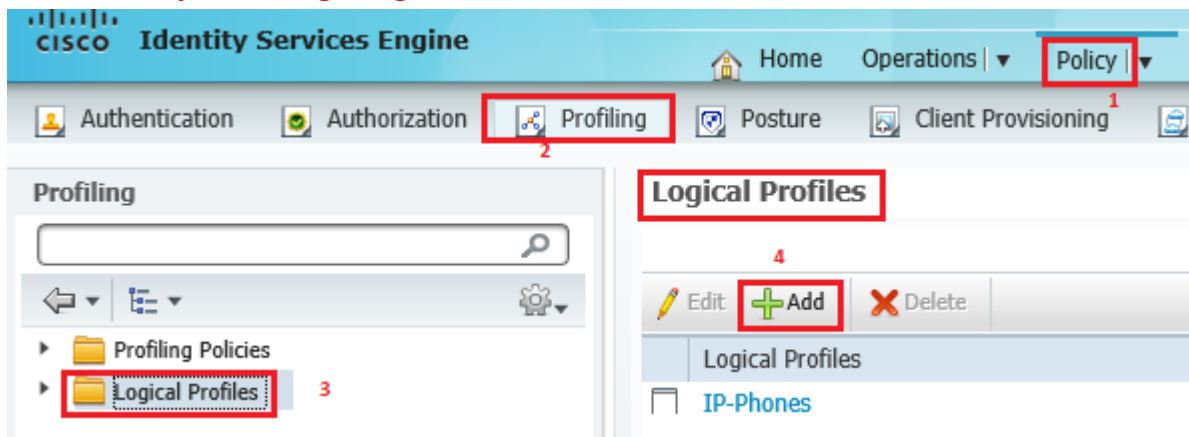
To modify per-profile, navigate to **Policy>Profiling>Profiling Policies** Click on any profile there is associated CoA Type

The screenshot shows the Cisco Identity Services Engine (ISE) Administration console. The navigation pane on the left has 'Profiling' highlighted, and 'Profiling Policies' is selected. The main content area is titled 'Profiler Policy List > Aastra-IP-Phone'. The 'Profiler Policy' configuration page is shown. It contains several fields: '\* Name:' with 'Aastra-IP-Phone'; 'Policy Enabled' checked; '\* Minimum Certainty Factor' set to '10'; '\* Exception Action' set to 'NONE'; '\* Network Scan (NMAP) Action' set to 'NONE'; 'Create an Identity Group for the policy' with 'No, use existing Identity Group hierarchy' selected; '\* Parent Policy' set to 'Aastra-Device'; and '\* Associated CoA Type' with a dropdown menu showing 'Global Settings', 'No CoA', 'Port Bounce', 'Reauth', and 'Global Settings'. The 'Global Settings' option is selected in the dropdown.

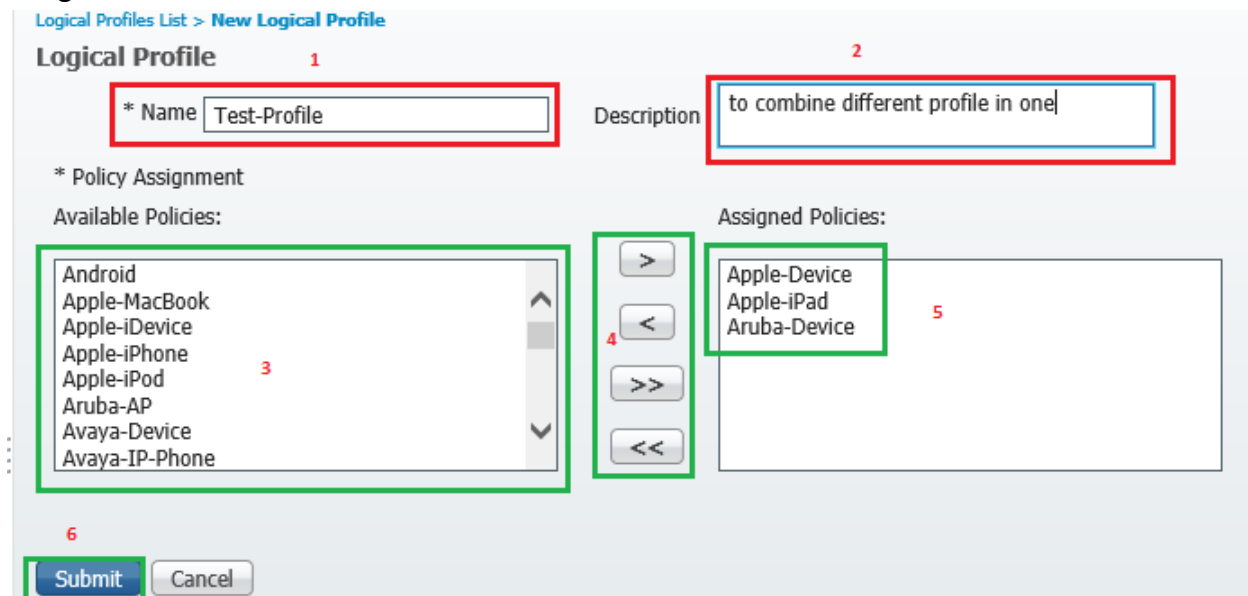
## Logical Profile:

- o Logical profile is a container for a category of profiles or associated profiles.
- o Logical profile administrator-created endpoint profiling groups and policies.
- o An endpoint profiling policy can be associated to multiple logical profiles.
- o Logical profile can be used in an authorization policy condition as well.
- o Logical profiles allow to group different type of dynamically profiled devices.

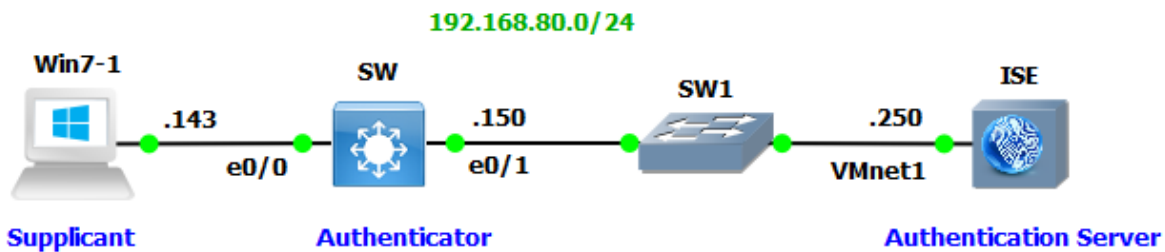
**Choose Policy > Profiling > Logical Profiles.**



Click Add. Enter a name and description for the new logical profile in the text boxes for Name and Description. Choose endpoint-profiling policies from the Available Policies to assign them in a logical profile. Click the right arrow to move the selected endpoint profiling policies to the Assigned Policies. Click Submit.



## Lab Topology:



### Dot1x Configuration

```
SW(config)#interface vlan 1
SW(config-if)#ip add 192.168.80.150 255.255.255.0
SW(config-if)#no shutdown
SW(config-if)#exit

SW(config)# aaa new-model
SW(config)# username admin password 123
SW(config)# radius server myserver
SW(config-radius-server)#address ipv4 192.168.80.250
SW(config-radius-server)#key test123

SW(config)# aaa authentication dot1x default group radius
SW(config)#aaa authorization network default group radius
SW(config)#aaa accounting dot1x default start-stop group radius
SW(config)#radius-server vsa send
SW(config)#radius-server vsa send accounting
SW(config)#radius-server attribute 6 on-for-login-auth
SW(config)#radius-server attribute 8 include-in-access-req
SW(config)#radius-server attribute 25 access-request include
SW(config)#ip device tracking
SW(config)# dot1x system-auth-control

SW(config)# interface e0/0
SW(config-if)# switchport host
SW(config-if)# switchport access vlan 1
SW(config-if)# authentication port-control auto
SW(config-if)# dot1x pae authenticator
SW(config-if)# authentication host-mode multi-host
SW(config-if)#authentication timer reauthenticate server
SW(config-if)#authentication event fail action next-method
SW(config-if)#authentication periodic
SW(config-if)#authentication violation shutdown

SW# show dot1x interface e0/0
SW# show dot1x all
```

## RADIUS Probe:

- o RADIUS Probe is one of the most common probes using by ISE.
- o ISE profile based on RADIUS attributes collected from RADIUS messages.
- o RADIUS probe also listens to CDP & DHCP attributes send in RADIUS accounting packets.
- o Such as User-Name, Calling-Station-Id, NAS-IP-Address, NAS-Port & Framed-IP-Address.
- o There is a lot of more information that can be pulled from the RADIUS attributes.

| Add or Remove Columns ▾ Refresh Reset Repeat Counts |         |              |          |                   |                       |
|-----------------------------------------------------|---------|--------------|----------|-------------------|-----------------------|
| Status                                              | Details | Repeat Count | Identity | Endpoint ID       | Endpoint Profile      |
| All ▾                                               |         |              |          |                   |                       |
| i2                                                  |         | 2            | admin1   | F0:6E:0B:12:34:56 | Microsoft-Workstation |
| i0                                                  |         |              | admin1   | F0:6E:0B:12:34:56 | Unknown               |

| Refresh + Add Trash ▾ Edit MDM Actions ▾ Refresh MDM Partner Endpoint Import ▾ Export ▾ |                   |             |                  |          |            |                   |                 |
|-----------------------------------------------------------------------------------------|-------------------|-------------|------------------|----------|------------|-------------------|-----------------|
| Endpoint Profile                                                                        | MAC Address       | Vendor(OUI) | Logical Profiles | Hostname | MDM Server | Device Identifier | IP Address      |
| ✕ Endpoint Prof                                                                         | MAC Address       |             |                  | Hostname | MDM Serv   | Device Id         | IP Address      |
| <input type="checkbox"/> Microsoft-Workstation                                          | F0:6E:0B:12:34:56 | UNKNOWN     |                  | WIN7     |            |                   | 192.168.80.1... |

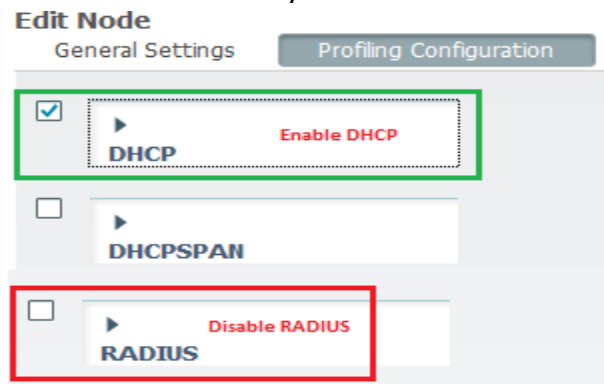
Looking at one of profiled hosts **Administration>Identity Management>Identities>Endpoints**, see the attributes that were collected:

|                          |                              |
|--------------------------|------------------------------|
| Called-Station-ID        | AA-BB-CC-00-01-00            |
| Calling-Station-ID       | F0-6E-0B-12-34-56            |
| DestinationIPAddress     | 192.168.80.250               |
| DestinationPort          | 1645                         |
| Device IP Address        | 192.168.80.150               |
| Device Type              | Device Type#All Device Types |
| DeviceRegistrationStatus | NotRegistered                |
| ElapsedDays              | 0                            |
| EndPointMACAddress       | F0-6E-0B-12-34-56            |
| EndPointPolicy           | Microsoft-Workstation        |
| EndPointProfilerServer   | ISE1.google.com              |
| EndPointSource           | RADIUS Probe                 |
| Framed-IP-Address        | 192.168.80.148               |
| IdentityGroup            | Workstation                  |

## DHCP Probe:

- o DHCP Probe collect DHCP request attribute from user, proxy & Helper.
- o Traffic forwarded to Cisco ISE with helper address or SPAN.
- o DHCP Probe using ip helper-address interface configuration command.
- o Cisco ISE will only use the incoming DHCP data to profile endpoints.

Navigate to **Administration > System > Deployment > Profiling Configuration** this time disable RADIUS Probe and only enable DHCP Probe.



Configure the switch to forward DHCP request to Cisco ISE.

```
SW(config)#interface vlan 1
SW(config-if)#ip helper-address 192.168.80.1
SW(config-if)#ip helper-address 192.168.80.250
```

Let us look again the profiled hosts **Administration > Identity Management > Identities > Endpoints**, see the attributes that were collected this time by DHCP.

|                        |                       |
|------------------------|-----------------------|
| EndPointPolicy         | Microsoft-Workstation |
| EndPointProfilerServer | ISE2.google.com       |
| EndPointSource         | DHCP Probe            |
| IdentityGroup          | Workstation           |
| MACAddress             | 00:0C:29:55:68:E0     |
| MatchedPolicy          | Microsoft-Workstation |
| OUI                    | VMware, Inc.          |
| PolicyVersion          | 0                     |
| StaticAssignment       | false                 |
| StaticGroupAssignment  | false                 |
| Total Certainty Factor | 30                    |
| UpdateTime             | 1562138211055         |
| chaddr                 | 00:0c:29:55:68:e0     |
| ciaddr                 | 0.0.0.0               |
| client-fqdn            | WIN7                  |

### SNMPQUERY Probe:

- o SNMP Query Probe send query packets known as SNMP GET Requests.
- o System Queries are periodic depending upon polling interval.
- o System Queries collect Bridge, IP CDP Cache Entry LLDP Local System Data.
- o Interface Queries are generated when RADIUS Accounting Starts.
- o Interface Queries are generated when SNMP detect linkup.

☒ **SNMPQUERY**

Retries

Timeout

EventTimeout

Description

### SNMP Trap Probe:

- o SNMP Trap receives information from configured NAD support MAC notification.
- o SNMP Trap receives info form configured NAD support linkup, linkdown & informs.
- o For SNMPTRAP to be functional, you must also enable the SNMPQUERY probe.
- o To make this feature functional, configure the NAD to send SNMP traps or informs.
- o SNMP Trap probe receives information from the specific network access devices.
- o When ports up or go down & endpoints disconnect from or connect to network.

☒ **SNMPTRAP**

Link Trap Query ☒

MAC Trap Query ☒

Interface

Port

Description

Go to **Administration > Network Resources > Network devices** click on network device in my case switch go down to **SNMP Settings** enable **SNMP Settings**, Choose version **2c** and type SNMP Community **123**. Click **Save** button to save the settings.

SNMP Settings

\* SNMP Version: 2c

\* SNMP RO Community: 123

SNMP Username:

Security Level:

Auth Protocol:

Auth Password:

Privacy Protocol:

Privacy Password:

\* Polling Interval: 600 seconds (Valid Range 600 to 86400 or zero)

Link Trap Query: ☒

MAC Trap Query: ☒

\* Originating Policy Services Node: Auto

#### Configuration of SNMP on Switch

S1(configure)# snmp-server community **123** rw

S1(configure)# snmp-server enable traps

S1(configure)# snmp-server host 192.168.8.250 **123**

Let us look again the profiled hosts **Administration>Identity Management> Identities> Endpoints**, see the attributes that were collected this time by **SNMPQuery**.

#### Attribute List

|                          |                   |
|--------------------------|-------------------|
| BYODRegistration         | Unknown           |
| DeviceRegistrationStatus | NotRegistered     |
| ElapsedDays              | 0                 |
| EndPointPolicy           | Cisco-Device      |
| EndPointProfilerServer   | ISE1.google.com   |
| EndPointSource           | SNMPQuery Probe   |
| IdentityGroup            | Profiled          |
| InactiveDays             | 0                 |
| MACAddress               | AA:BB:CC:80:01:00 |